Online Activism and Dyadic Representation

Abstract

When politicians lack information on their constituents' preferences, dyadic representation is likely to suffer. The internet has made it significantly easier for citizens to communicate their political preferences, meaning that the expansion of online political activity has the potential to strengthen the representational link between politicians and voters. However, we know little about the effects of online advocacy on politicians' behaviour. I use new data from an electronic petition (e-petition) system in the UK to examine whether Members of Parliament (MPs) are responsive to constituent activism that occurs online. I show that support for an e-petition amongst an MP's constituents increases the probability that the MP participates in parliamentary debate relevant to the petition, even when controlling for participation on similar topics in non-petition debates. However, online advocacy is effective only on issues that are not subject to strong party discipline, and only for MPs facing competitive electoral environments.

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Many normative theories of democratic representation suggest that, when viewed as delegates, politicians should advance the preferences of their constituents in the political sphere (Pitkin 1967). In single-member district (SMD) electoral systems, a key concern is whether legislators' actions reflect the specific views of their districts – a concept usually described as "dyadic" representation (Weissberg 1978). Indeed, the idea that SMD-systems incentivise responsiveness to local concerns is commonly cited as one of their central virtues (Norris 2001, Carey & Hix 2011). However, a central dilemma for reelection-seeking politicians is that acquiring information on the preferences of voters can be difficult and costly. This can have pernicious effects, for even when politicians have incentives to adapt their behaviour to reflect constituents' wishes, a paucity of information will mean that politicians' knowledge of constituent preferences may be biased (Kingdon 1967, Broockman & Skovron 2018); responsiveness will be limited (Stimson et al. 1995); and that the quality of democratic representation is likely to suffer (Mansbridge 2003, 518). Put bluntly, it is hard to do what the voters want if you do not know what it is that the voters want you to do.

The growth of online political activity therefore represents an important change to the information environment faced by politicians. By lowering the costs of acquiring political knowledge and giving voters new mechanisms for communicating with their representatives, the internet has the potential to improve the informational link between politicians and voters. For some, this promises to strengthen representation and "enrich democracy." (Coleman & Blumler 2009, 11) In an attempt to capitalise on these benefits, government-sponsored electronic petition (e-petition) systems have been adopted in many countries in recent years.[1] E-petitions are seen as a new tool for galvanising support for policy (Hersh & Schaffner 2018); a gateway to other forms of participation (Kennedy Stewart et al. 2013); and, critically, as a method for strengthening the representational bond between representatives and their constituents (Hough 2012, Bochel 2012, Procedure Committee 2014). Empirical evidence on the consequences of these initiatives for democratic representation has, however, been

[1] E-petition schemes operate in the UK (https://petition.parliament.uk), the US (https://petitions.whitehouse.gov), Germany (https://epetitionen.bundestag.de), Canada (https://petitions.ourcommons.ca), the Ukraine (https://petitions.com.ua), as well as in the European Parliament (https://petiport.secure.europarl.europa.eu/petitions) and in many local government settings (Bochel 2012, Hough 2012).

noticeably absent from the political science literature.

I investigate the effects of online advocacy on dyadic representation by examining data on e-petitions and legislative behaviour in the UK House of Commons. I argue that e-petitions represent a valuable source of information on constituent preferences for reelection-seeking Members of Parliament (MPs). Legislators typically learn about constituent preferences through imperfect signals – emails, letters, social media, etc – which makes it difficult to form correct judgements of constituency opinion, and potentially inhibits legislator responsiveness. The petition data I study, by contrast, provides legislators with a natural metric for measuring the strength of district opinion across a variety of issues. As petition signature counts are available at the parliamentary constituency level, MPs can use this information to guide their legislative actions in ways that are likely to be electorally rewarding, and in so doing may improve the representation of constituent interests in the policy process.

Methodologically, this data also ameliorates problems in existing studies of dyadic representation in parliamentary systems, which either rely on demographic proxies for measuring district opinion (Soroka et al. 2009, Blidook & Kerby 2011), or are limited to evaluating the dyadic relationship on a small number of high-profile issues that are covered in national opinion polls (Hanretty et al. 2016). E-petitions offer political scientists (as well as legislators) clear, quantitative and district-specific information about the relative salience of a wide variety of political issues.

Further, most studies evaluate whether legislators cater to district preferences by adapting their roll-call *voting* behaviour (Miller & Stokes 1963, Erikson 1978, Kastellec et al. 2010, Krimmel et al. 2016), but strong party discipline means that dyadic voting is unlikely to manifest in most parliamentary systems (Powell Jr 2004, 283-285). However, the UK petition system suggests an alternative setting for dyadic representation: in parliamentary debates. A critical feature of this system is that e-petitions receiving more than 100,000 signatures are considered for debate in parliament. In addition to exhibiting more variation than legislators' voting decisions, an MP's debate participation – the main dependent variable of interest here – constitutes an important representative act, as making a speech in a debate of great local interest allows an MP to literally 'speak up' for her constituents' interests. I therefore expect

MPs to be more likely to participate in debates on petitions marked by high levels of local support.

Identifying these effects, however, is complicated by the fact that if an MP already has a strong sense of which issues are important to her constituents, she might participate in a given petition debate even in the absence of the information provided by the relevant e-petition. If voters use elections to select representatives who share their policy concerns, or pressure legislators to take action on certain issues through other mechanisms, then the relationship between petition signatures and debate behaviour may be confounded. These alternative mechanisms, however, are likely to result in predictable behaviour across *many* parliamentary debates, not just those devoted to petitions. An MP from an agricultural constituency, for example, will likely receive many signatures on agricultural petitions, will probably attend the relevant petition debates, but will probably also attend *non-petition* agricultural debates. I make use of this fact in an empirical strategy where, by estimating each MP's tendency to speak in many debates on certain topics, I am able to examine their responsiveness to e-petitions net of these more general representational behaviours. Combining statistical topic models with a multilevel random-effects model, I show that MPs' responsiveness to constituent activism in petition debates differs from their behaviour in counterfactual non-petition debates that are otherwise very similar in terms of topical content.

The results suggest that, in general, when many of an MP's constituents sign an e-petition, that MP is substantially more likely to *participate* in the relevant parliamentary debate. I also provide evidence that MPs are more likely to use their speeches to *express agreement* for e-petitions marked by stronger local support. However, party discipline clearly affects responsiveness: MPs in government positions are less responsive to petition support than backbench MPs, and when petitions address issues which are central to existing partisan divisions, signature rates are inconsequential. In addition, responsiveness also varies according to electoral considerations, and I document stronger effects for MPs who represent competitive constituencies than for MPs in 'safe' seats. Overall, while policymakers argue that online petitions have the potential to "bring about a significant enhancement" of the relationship between politicians and voters (Procedure Committee 2014, 30-31), the findings here suggest

that the effects of online activism on dyadic representation will likely be constrained by existing institutional structures, and will only affect the legislative behaviour of certain subsets of MPs.

Information and Dyadic Representation in Parliamentary Systems

Positive accounts of representation suggest that reelection-seeking politicians will tailor their behaviour to satisfy the wishes voters, either by taking positions that align with district preferences, or by allocating attention to the most salient district-specific issues (Downs 1957, Mayhew 1974, Cain et al. 1984, Hall 1996). Information is a central component of these theories (Mansbridge 2003, 516–520), as legislators are expected to infer the preferences of their constituents, and use this knowledge to guide their legislative decisions. However, legislators will only occasionally have access to high quality information – such as district-specific opinion polls – to make inferences about the opinions of their constituents on particular policy issues. More commonly, legislators must rely on heuristics to evaluate the strength of opinion towards a specific policy in their constituencies: how much correspondence has the legislative office received? How prominent is the issue in local media? It is therefore unsurprising that politicians invest significant resources in gathering additional information about constituent preferences (Maestas 2003), and that even small changes in this information can translate into meaningful changes in legislative behaviour (Butler et al. 2017, Bergan & Cole 2015).

For legislators, the value of online political expression, and of e-petitions in particular, resides in the additional signal it offers about constituents' preferences. First, e-petitions provide a clear quantitative measure of the relative importance of different issues. The number of signatures a petition receives can be easily compared to the signature rate on another petition, allowing legislators to make simple comparisons of salience across issues. Second, e-petitions tend to focus on a more diverse set of topics than those that traditionally dominate national debate (Hersh & Schaffner 2018), and which also contrasts with the limited range of issues considered by national opinion polls. Third, e-petition signature counts are typically available at low geographic levels of aggregation.[2] This is critical, as locally-disaggregated information allows legislators to respond to the specific preferences of their constituents, which

[2]Signature counts are available at the constituency-level in the UK (Procedure Committee 2014, 48) and at the zip-code level in the US (Hersh & Schaffner 2018).

may differ from those of the population at large.

The information provided by e-petitions may not be *better* than the signal produced by more traditional forms of interaction with constituents, but it is likely that it will at least be *different*. For example, the people who participate in online political activities are different from those who participate offline (Marien et al. 2010), and this holds for the subset of people who sign e-petitions.[3] Regardless of whether it is an accurate representation of average voter preferences, if the information provided by petitions differs from legislators' priors, we should expect legislators to update their beliefs and change their behaviour in ways that are more consistent with constituent preferences as they are expressed in the petition. As Bochel (2012, 798) argues, e-petitions may facilitate representation "by providing a mechanism to enable the public to express their views to those in elected representative institutions."

However, in parliamentary systems these informational effects will be mitigated by countervailing incentives to toe the party line. An inherent tension in SMD parliamentary systems is that "electoral rules create incentives for a personal vote and legislative rules deny representatives the opportunity to establish it." (Cain et al. 1984, 217) In particular, legislators in parliamentary systems will be less able to use roll-call votes as a tool for demonstrating responsiveness than is the case for legislators in presidential systems (Carey 2007). Therefore, the provision of constituency-specific information may generate significant incentives for dyadic behaviour, but the party-discipline constraints in parliamentary systems may mean that legislators are unable to respond to such information.

Whether and under what conditions e-petitions affect legislative behaviour is ultimately an empirical question, but the academic literature has so far largely been limited to descriptive work on the e-petitioning process (Jungherr & Jürgens 2010, Lindner & Riehm 2011, Escher & Riehm 2017, Yasseri et al. 2017, Hersh & Schaffner 2018). While useful, this literature does not study the effectiveness of these systems on democratic representation. In the next section, I outline the system in the UK, and derive expectations for the effects of e-petitions on behaviour in the House of Commons.

[3]See e.g. Lindner & Riehm (2011), and appendix section S1 for evidence from the UK.

E-petitions and parliamentary debate in the UK

With more than 20 million signatures recorded between 2015 and 2016 (Petitions Committee 2016), the UK e-petition system has proved to be a popular new form of democratic engagement. Petition-signing is the second most popular form of political activity (after voting) in the UK and nearly 30% of adults report signing an e-petition in the past year (Hansard Society 2018). E-petitions require just five personally collected signatures before they are made publicly available on a dedicated website where anyone can sign by providing their name, email address and postcode. As petitioners provide their postcode, it is possible to calculate the number of people signing a given petition from a given constituency. The parliamentary Petitions Committee makes the constituency-level signature-counts available online, and also – for any petition that receives over 100 thousand signatures – provides this information directly to MPs.

How likely are MPs to respond to this information? First, signature counts are a clear source of information about the salience that different constituencies attribute to different issues. Constituency-specific opinion polls are rare in the UK (Wring et al. 2016) and if MPs can only approximate the issue priorities of their constituents from other sources, then the number of signatures for a given petition might provide valuable information that MPs can use to update their beliefs about the best ways to allocate their time to different issue areas. Furthermore, the search costs for acquiring this information are small: for debated petitions, the Petitions Committee circulates a spreadsheet of constituency signature counts directly to MPs.

Second, e-petitions also address a wide variety of issues – both salient and non-salient – on which it is unlikely that MPs will always have strong prior understanding of constituent preferences. For instance, the petition calling on the government to "Stop allowing immigrants into the UK"is typical of the type of issue that might feature in UK election campaigns. By contrast, the petition that lobbied the government to "Give the Meningitis B vaccine to ALL children" addressed an issue – not typically central to UK political debate – where MPs were likely to have weaker intuitions about the importance of the issue to their constituents. E-petitions may therefore also be informative about constituent opinion on issues that have not

previously been a part of the political agenda.

Third, e-petitions are formally integrated within the political process in Westminster and MPs have a clear setting for responding to petitioners' concerns. In particular, any petition that receives over 100,000 signatures nationally is considered for debate in parliament. Although the Petitions Committee has some discretion over which debates are scheduled, almost all petitions crossing the 100,000 threshold (as well as some slightly below it) have been debated.[4] These debates are attended by members of all political parties as well as the government minister responsible for the relevant department.

How might we examine responsiveness in the context of these debates? Most studies of dyadic representation focus on the congruence between constituency opinion and legislators' *voting* behaviour, but there is no substantive vote at the end of petition debates. One strategy I pursue below is to focus on the subset of legislators who participate in petition debates, and examine whether MPs whose constituents are more supportive of a petition are more likely to voice their approval of the petition in debate.

However, focussing only on those who participate obscures the fact that the decision to participate in a given debate can itself be a representational act. Politicians may respond to new information about constituent preferences not by changing their policy positions (as revealed by the votes they cast), but rather by shifting their legislative efforts towards issues that their constituents find most salient. As Hall (1996, 58) suggests, constituency influence should not only operate on legislators' voting behaviour "but on the intensities that they reveal in their decisions about when and to what extent they will participate in particular matters". By allocating more of their time to the most pressing district issues, politicians demonstrate their willingness to address constituent concerns.

Legislative debate is a useful arena for pursing this type of representational strategy, as parliamentary speeches are publicly visible acts that are easy to publicise to constituents. Parliamentary debates are widely covered in the UK media, and both voters and MPs consider participating in debate to be one of the most important roles that an MP performs (Campbell & Lovenduski 2014). Despite this, the average debate in the Commons is attended by only 12

[4]Table S2 in the appendix lists the 56 debates held between June 2015 and May 2017.

MPs.[5] Speaking in a debate, then, is a credible signal to constituents of the importance an MP assigns to a given issue in parliament. Standing up and making a speech in a debate devoted to a petition which many of your constituents have signed allows the MPs to demonstrate that she has heard local concerns, and has given voice to those concerns in parliament. Additionally, speechmaking is less subject to control by party elites than the votes cast in roll-calls (Proksch & Slapin 2012). If strong party discipline in parliamentary systems does mitigate incentives for dyadic representation, the fact that petition debates are not subject to roll-call votes suggests that MPs may be able to respond to the preferences of their constituents without challenging the party line.

In sum, the UK e-petition system provides MPs with clear information regarding constituent issue priorities, and a formal mechanism for responding to that information that is not generally subject to strict party discipline. I therefore expect that the probability that an MP participates in a petition debate will increase with the number of signatures on the relevant petition from the MP's constituency.

Hypothesis 1: *MPs are more likely to participate in relevant debates for petitions signed by many of their constituents than for petitions signed by few of their constituents.*

However, these effects are likely to vary across different petition debates. Some petitions address issues which are not typically the subject of political debate between parties in the UK, while other petitions address issues that have clear party-based dividing lines.[6] Petitions that address non-partisan issues are likely to have larger effects on legislative behaviour. First, MPs will have more diffuse beliefs about constituent preferences on issues that are not currently a major feature of the political agenda. Second, issues that are not connected to the main dimension of party competition may be less subject to party discipline than other issues, as parties will not see such issues as central to the party 'brand' (Cox & McCubbins 1993).

The UK system provides a convenient categorisation of petition debates that can be used to explore this expected heterogeneity. For most petitions, debates are held in a smaller ante-

[5] Author's own calculation.

[6] For example, while the petition entitled *Stop the Scathing Cuts to the Police Budget* clearly addressed a typical party-political issue, the petition entitled *Give Status to Police Dogs and Horses as 'Police Officers'* focused on an issue that was largely absent from the political agenda.

chamber called Westminster Hall. However, when a petition addresses an issue that is already scheduled for discussion in main chamber of the House of Commons, the petition is 'tagged' to the debate: MPs are informed that there is a petition that is relevant to the debate and the petition is listed on the parliamentary order paper for that day. I expect that the effects of e-petitions on debate behaviour will be weaker for these 'tagged' debates, as these petitions focus on issues that already represent key dividing points between parties.

Hypothesis 2: *The relationship between petition signatures and participation will be weaker for petitions debated in the* Commons' Chamber *than for those debated in* Westminster Hall.

Similarly, an MP's incentives to respond to petitioners' demands are likely to vary as a function of the electoral environment the MP faces. Politicians who win office in districts with small electoral margins have greater incentive to respond to constituents than politicians who win office by large margins (MacRae 1952), and the dyadic connection is indeed stronger for legislators in more competitive districts (Griffin 2006). I expect the same dynamics to operate in the UK: MPs in marginal seats will be more responsive to petition signatures than MPs in safe seats.

Hypothesis 3: *The relationship between petition signatures and participation will be stronger for MPs in more electorally competitive constituencies.*

Furthermore, MPs who hold government positions are more likely to be responsive to demands from party leaders than are backbench MPs (Benedetto & Hix 2007), and government MPs are formally responsible for attending certain debates in the House (those pertaining to the jurisdiction of their ministries) and so have less freedom to choose which debates to attend. Accordingly, the relationship between petition signatures and participation should be weaker for government MPs than for backbench MPs.

Hypothesis 4: *The relationship between petition signatures and participation will be weaker for MPs who are members of the government.*

Finally, MPs who have recently entered parliament, on the other hand, are likely to be more eager to establish a dyadic connection with their constituents than their more experienced

colleagues. New MPs have had less time to learn about constituent issue priorities, making the information from petitions more valuable. In addition, new MPs are significantly less time constrained, as they are less likely to hold front bench positions or to serve on time-intensive select committees. New MPs are also less likely to have established a reputation for standing up for their constituents, and so should be more willing to put on record their support for issues that have attracted significant local support.

Hypothesis 5: *The relationship between petition signatures and participation will be stronger for MPs who have recently entered parliament.*

Data

I use data on 56 petitions debated between the introduction of the current e-petition system in June 2015 and the end of the parliamentary term in May 2017, which I link to debate transcripts pertaining to each petition. The Petitions Committee will occasionally group two petitions in the same debate when they address very similar issues, and so the main data of interest come from 46 debates. As discussed below, I also require debate transcripts for all *non-petition* debates during the same parliamentary term. The total corpus consists of 2525 debates and close to 300,000 speeches.[7]

To construct the independent variable, I take the population normalised *constituency*-level signature counts for a given petition, and divide by the population normalised *national*-level signature counts for that petition.[8] The signature rate for MP i in debate d is given by:

$$\text{Signature rate}_{id} = \frac{\text{Constituency Signatures Per Capita}_{id}}{\text{National Signatures Per Capita}_{d}} - 1 \tag{1}$$

where I subtract 1 so that for all non-petition debates the signature rate can be set to 0 for all MPs, meaning that the non-petition debates will not contribute to the estimation of coefficients relating to the signature rate variable.[9]

[7] Petition data was collected from www.petition.parliament.uk and speech data from www.hansard.parliament.uk.

[8] When more than one petition is discussed, I sum the constituency signature counts across the petitions.

[9] An alternative formulation would be to take the log of the ratio, but this is unattractive because there are constituencies with zero signatures for a given petition in the data.

Normalising the counts by population size accounts for the fact that some constituencies are larger than others. Dividing by the national signature rate accounts for the fact that some petitions are more popular than others. Equation 1 therefore implies that an MP's signature rate will be larger when her constituents sign a petition at a greater rate than constituents from other constituencies. For example, when equation 1 is equal to 0, constituency i's signature rate for petition d is equal to the national rate of signatures for petition d. When equation 1 is equal to 1, the constituency signature rate is double that of the national rate, and when it is equal to -0.5 the constituency signature rate is half the national rate.

The measure assumes that MPs are more likely to participate in a petition debate if their constituency is high-ranked relative to *other constituencies* in terms of signatures. Another plausible assumption is that MPs are more likely to participate in debate on a given petition if they receive more constituency signatures than received for *other petitions*. The former of these possibilities seems most likely. The Petitions Committee provides MPs with constituency signature counts in a spreadsheet format on a petition by petition basis. Making comparisons relative to other constituencies on a given petition is straightforward: an MP (or her staff) need simply look up this information in the spreadsheet. Making comparisons to past petitions is more difficult: an MP would need to analyse historical data to establish whether constituents found the current petition more salient than previous petitions.

In the main analysis, the dependent variable measures whether MP i participates in debate d or not. An MP is coded as participating when she makes a speech in the debate ($y_{id} = 1$), and zero otherwise. Of the 1.6 million MP-debate observations from both petition and non-petition debates, approximately 3% indicate participation. I also consider the effects of signatures on petition *agreement*. Restricting the sample to petition debates, an MP is coded as agreeing with a petition if she expressed overt support for the petition proposal in her speech.[10] Conditional on participating in a petition debate, approximately 55% of MPs agreed with the aims of the relevant petition.

Figure 1 shows the probability of debate participation (panel A) and petition agreement (panel B) over the range of the signature rate variable for the 56 petition debates.[11] Grey

[10] Full details are given below.

[11] Very few observations have a signature rate greater than 7 and in these plots I restrict to values between

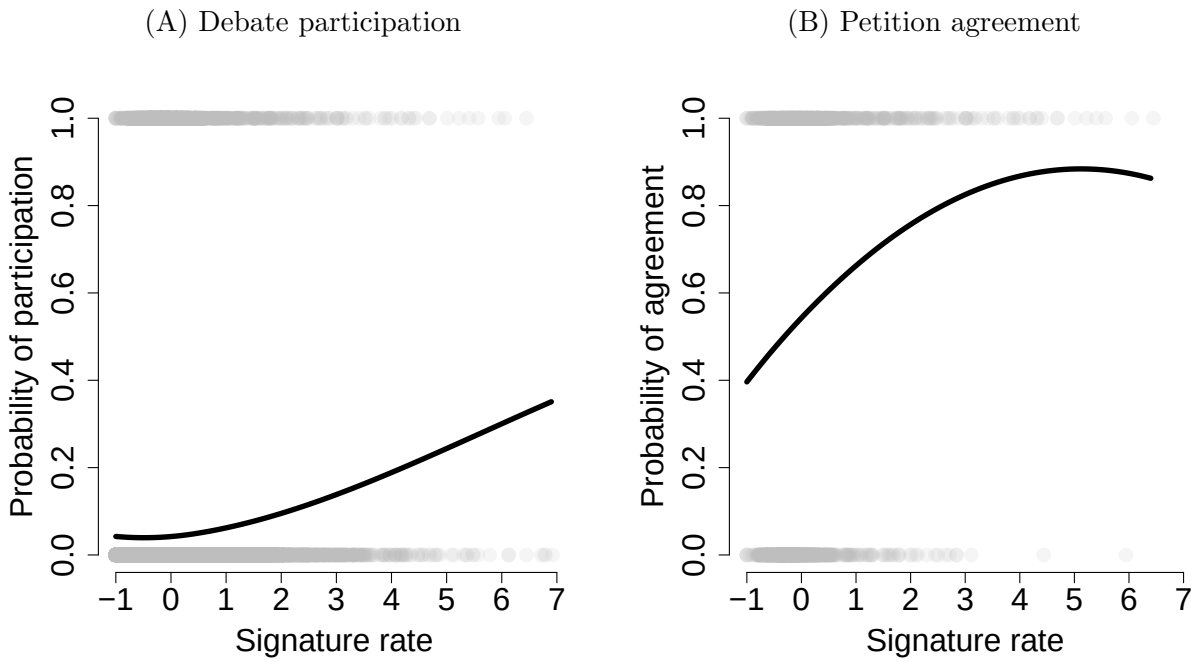

Figure 1: **Petition signatures predict debate participation and petition agreement** Panel A: as the signature rate increases, MPs are more likely to participate in debate. Panel B: conditional on participation, an MP is more likely to express agreement with a petition as the signature rate increases.

points are the raw data and the black lines are loess curves. The plots illustrate that, in the raw data at least, there is a clear positive relationship between petition signatures and both outcome variables. Bivariate logistic regressions imply that increasing the signature rate by one unit increases the odds of debate participation by 50% (z = 12.06, assuming independence) and, conditional on an MP participating in the relevant debate, the odds of expressing agreement with the petition by 61% (z = 6.24, assuming independence).

Empirical strategy

Simple comparisons of participation between MPs from constituencies with low and high signature rates are likely to be confounded in a number of ways. First, some constituencies will have higher signature counts, on average, than other constituencies.[12] If MPs from these constituencies are also more active in debates *in general* than other MPs, then any association

-1 and 7.

[12]See appendix section S1.

between signatures and participation might reflect these general patterns. To overcome this problem I estimate MP-specific intercepts that account for fixed tendencies for some MPs to participate in debate more than others, and which imply that identification of the signature rate effect relies only on within-MP variation.

Second, some petitions are clearly more salient than others. For example, the "EU Referendum Rules triggering a 2nd EU Referendum" petition received nearly 2 million signatures nationally and was subject to extensive media interest. Debates on such popular petitions are likely to be better attended on average than other petition debates, and MPs' participation is likely due to the high-profile nature of the debate in general, rather than because of the advocacy of their constituents in particular. I account for this concern in two ways. First, as explained above, the signature rate variable increases only when a constituency provides more signatures for a given petition relative to the total national signature rate for that petition. Second, I therefore also include debate-specific intercepts in all models that account for commons shocks to participation that affect all MPs in a given debate.

A more subtle inferential concern, however, is that debate participation may not result from constituent activism via e-petitions, but may instead reflect MPs' pre-existing tendencies to devote attention to the types of issue that attract the attention of their constituents. For example, MPs representing rural areas are likely to actively participate in debates that pertain to farming issues, even if such debates are not linked to e-petitions that deal with farming. Likewise, MPs from constituencies with high numbers of university students are likely to attend debates about education, even in the absence of a petition calling on them to do so. Even if MPs attend debates linked to locally popular petitions, this may be because they already acting as 'good' representatives, and not changing their behaviour in response to the petition.

Figure 2 gives two examples of this type of issue-specific confounding. In both panels, the x-axis indicates the signature rate for an example petition. Panel A depicts a petition calling for improvements to the National Health Service (NHS). Panel B shows a petition which aimed to restore an educational grant which had recently been cut by the government. The y-axis measures whether an MP participated in any *non-petition* debates that related to these issues,

(A) Participation in non-petition NHS debates (B) Participation in non-petition student debates

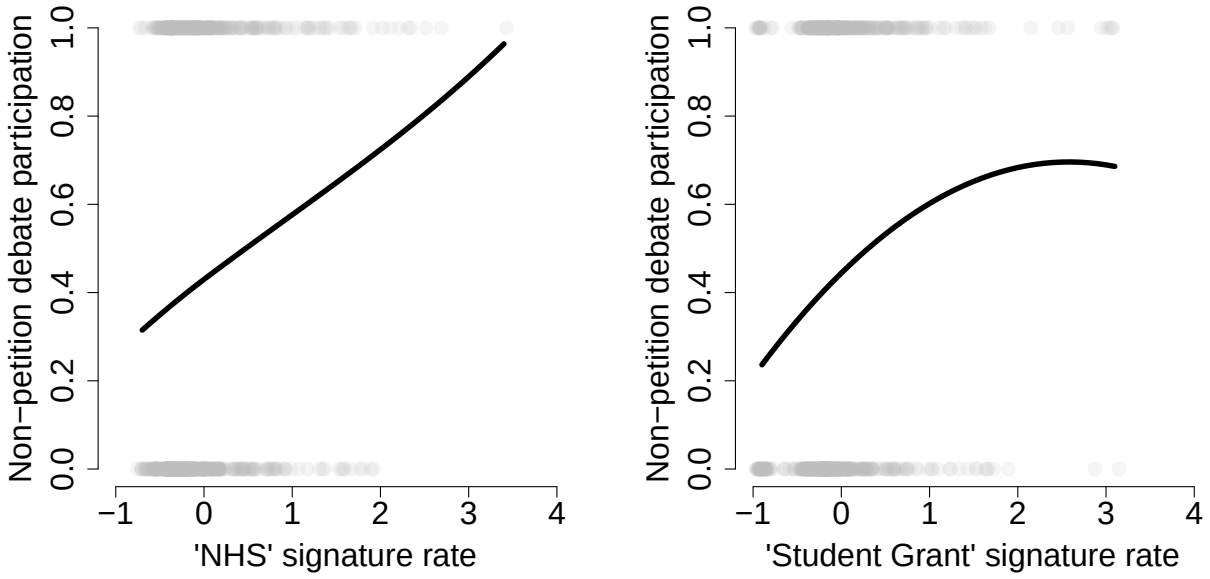

Figure 2: **Petition signatures predict participation in related *non-petition* debates** Panel A: MPs who received more 'NHS' petition signatures are more likely to participate in non-petition healthcare debates. Panel B: MPs who received more 'Student Grant' petition signatures are are more likely to have participated in non-petition student debates.

and which were held *before* the relevant petition was discussed in parliament. For panel A, an MP is counted as participating if she spoke in any debate which occurred prior to the 'NHS' petition, and for which the title included the word 'NHS'. For panel B, I measure which MPs spoke in debates that occurred prior to the 'Student Grant' petition where the debate-title included the words 'Student', 'College', or 'School'. In both cases, petition signatures clearly predict MP participation in prior *non-petition* debates that relate to the same issues as those raised in petitions.

Consequently, the relationship between petition signatures and participation is likely confounded by the fact that MPs will tend to speak frequently about issues which are of concern to their constituents, even in the absence of the information that they glean from the petition. This implies that we should account for an MP's general tendency to participate in debates that relate to the issues of interest that are raised in petitions. In essence, we would like to be able to compare an MP's participation on a given petition debate to a counterfactual debate

for which the MP was not provided with petition information. Although we cannot observe this counterfactual, we can make comparisons between petition and non-petition debates that deal with similar underlying topics. The empirical challenge is to design a procedure that maps petition debates to other debates in terms of their shared topical content, and quantifies the tendency of each MP to participate in debates on those topics.

First, I use statistical topic models to characterise the topics that feature in parliamentary debate, and to estimate the proportion of each debate that pertains to each topic. Second, I incorporate these debate-specific topic proportions into the participation model by including coefficients for each MP-topic combination. These effects describe the relative tendency for a given MP to participate on a given topic (for both petition and non-petition debates), and allow me to control for the possibility that some MPs will be more prone than others to discuss certain topics. If an MP tends to participate in many debates about healthcare, for example, then that MP will have a large 'healthcare' topic random-effect. To the extent that an MP's participation in debates relevant to specific topics correlates with her constituents' signatures of petitions that are also relevant to those topics, this strategy will lead to more conservative estimates of the effects of petition signatures.

The full model specification represents a conservative strategy for estimating the effect of signatures on participation. MP-specific intercepts rule out omitted variable bias that stems from MP characteristics that are time invariant; debate-specific intercepts control for confounding factors that relate to the aggregate popularity of each debate; and the MP-topic coefficients account for any factors that might drive MP and constituent attention to particular topics.[13]

Model specification

To implement the topic-control strategy, I estimate statistical topic models using the texts of all speeches during the study period. In particular, I concatenate all speeches within each debate into a common text, and then estimate Correlated Topic Models (Blei & Lafferty 2005)

[13] I consider further threats to inference in the penultimate section.

models for all D parliamentary debates, including petition debates.[14] Like all topic models, the CTM assumes that the frequency with which terms co-occur within different debates gives information about the 'topics' that feature in those debates. The key output of the model is λ, which is a $K * D$ matrix, where λ_d describes a given debate as a vector of topic proportions, and λ_{kd} gives the proportion of debate d devoted to topic k. Note that I use the texts of *all* debates – both petition-related and non-petition-related – to estimate the topic models. This is essential, as the resulting λ matrices therefore provide a common topic space which we can use to compare petition debates to regular debates.

The CTM requires that the we choose a number of topics, K, to estimate from the data. Because there is no *a priori* reason to prefer any particular value for K, I estimate several different models, while varying the number of topics: $K \in 10, 20, ..., 50$. I present results from all 5 topic model specifications in appendix figure S2, but appendix section S2 shows that the model with 20 topics best predicts MP debate participation and I focus on this results from this model for the remainder of the paper.

With the topic estimates in hand, for individuals $i, ..., N$, debates $d, ..., D$, topics $k, ..., K$, and locations $l \in \{1 = \text{Commons Chamber}, 2 = \text{Westminster Hall}\}$, I model debate participation in a hierarchical logistic regression of the following form:

$$\text{logit}(y_{id(l)}) = \alpha_0 + \alpha_i + \delta_d + \sum_{k=1}^{K} \beta_{ik}\lambda_{kd} + \gamma_{i,l} * \text{Signature rate}_{id(l)} \tag{2}$$

where $y_{id(l)}$ measures whether an MP participated in a given debate and Signature rate$_{id}$ is defined in equation 1. α_i and δ_d are intercepts for MPs and debates, respectively, α_0 is a grand-intercept term, and β is a matrix of MP-topic coefficients. Finally, γ is a matrix of MP-signature coefficients.

The primary quantities of interest are the γ_i coefficients, which describe the relationship between the signature rate and debate participation for each MP. The l subscript indicates that I estimate two signature rate coefficients per MP – one for debates in the Commons' Chamber, and one for debates in Westminster hall – which allows me to capture the expected

[14] I remove all punctuation and numbers from the texts, reduce words to their stems, and remove any tokens that appear in fewer than 2.5% of all texts. I implement the CTM as the null specification of the structural topic model Roberts et al. (2013)

heterogeneity across debate settings in line with hypothesis 2. As these coefficients are estimated for each MP, I use only within-MP variation in debate participation to identify the effect of petition signatures. Recall also that as Signature rate$_{id} = 0$ for all non-petition debates, it is only petition debates that contribute to the likelihood function via the γ parameters.

As outlined above, I also expect the effect of signatures to vary across different MP types, and accordingly I model the γ coefficients in a second-level model of the following form:

$$\gamma_{i,l} \sim N(\varphi_{0_l} + \varphi_{\text{New MP}_l} * \text{New MP}_i + \varphi_{\text{Government}_l} * \text{Government}_i + \varphi_{\text{Margin}_l} * \text{Margin}_i, \sigma^2_{\gamma,l}) \quad (3)$$

where Government$_i$ is a binary variable for whether a given MP held a government position during the 2015-2017 parliament, New MP$_i$ is an indicator for whether the MP was new to the House in 2015, and Margin$_i$ is the electoral margin of victory for MP i in 2015. The coefficients associated with these second-level variables will tell us how (on the logit scale) the γ_i coefficients vary, on average, according to the attributes of the MP. The model therefore provides two major benefits: first, it includes MP-specific topic effects, which mitigate a broad array of confounding factors; and second, it provides a rich framework for evaluating how the effects of e-petitions vary as a function of debate- and MP-level characteristics.[15]

Estimation is complicated by the fact that we have lots of data (≈ 1.6 million observations), lots of models ($K \in 10, 20, ..., 50$), and lots of parameters per model.[16] Traditional Bayesian inference approaches such as MCMC would be computationally impossible here, and so I instead use a variational inference approach (Kucukelbir et al. 2017) to approximate the posterior distribution, which I implement in Stan (Carpenter et al. 2016).[17]

[15] Although of less substantive interest, I also include the same covariates in a second-level model for the MP-intercepts (α_i), and model debate effects as a function of whether a debate is associated to a petition or not. The model is completed by uninformative prior distributions over the β_{ik} coefficients, and the variance parameters:

$$\begin{aligned} \alpha_i \sim N(\mu_0 + \mu_{\text{New MP}} * \text{New MP}_i + \mu_{\text{Government}} * \text{Government}_i + \mu_{\text{Margin}} * \text{Margin}_i, \sigma^2_\alpha) \quad (4) \\ \delta_d \sim N(\phi_0 + \phi_{\text{Petition Debate}} * \text{Petition Debate}_d, \sigma^2_\delta) \\ \beta_{i,k} \sim N(0, \sigma^2_{\beta_k}) \\ \sigma_\alpha, \sigma_\delta, \sigma_\gamma, \sigma_{\beta_k} \sim N(0, 2) \end{aligned}$$

[16] A model with 10 topics includes over 10 thousand parameters, and a model with 50 topics includes nearly 40 thousand.

[17] Variational inference is orders of magnitude faster than state-of-the-art MCMC algorithms, and is guaranteed to provide the correct expected values of parameters. However, it will also sometimes understate the

In appendix section S2 I provide extensive validation of the topic control strategy. I show that this approach improves our ability to predict MP debate participation in general; associates MPs to topics in ways that are consistent with intuitions about British politics; and, crucially, links petition-debates to counterfactual non-petition-debates that are very similar in terms of their topical content. With these validity checks in hand, I turn now to the results.

Results

Table 1 presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equation 3.[18] Model 1 includes only the γ parameters from equation 2 and the associated second-level parameters from equation 3. Models 2 and 3 include intercepts for MPs (α) and debates (δ), respectively, and both sets of intercepts are included in model 4. Model 5 additionally includes the MP-topic coefficients (β), and model 6 includes the full set of effects and presents the second-level signature-rate coefficients for both the Commons (left column) and Westminster Hall (right column). Although there are some small differences across models, conditioning on MP-effects, debate-effects, and MP-topic-effects makes little substantive difference to the estimated effects of petition signatures. That the signature-rate effects differ little after the inclusion of these controls is reassuring, and considerably strengthens the inferences drawn here.

Consider the baseline signature effect in each model, φ_0, which represents the effects of the petition signature rate for an MP in a perfectly competitive constituency (where Margin = 0), who is not new to the House and who does not hold a government position. Across all specifications, this baseline effect is positive, and precisely estimated. Based on the pooled effect in model 5, increasing the signature rate by one unit is associated with a 28% [19%, 37%] increase in the odds of debate participation for an MP of this type. Model 6 indicates that the pooled estimate masks important heterogeneity: for debates held in Westminster Hall, a one-unit signature-rate increase is associated with a 38% [25%, 53%] increase in participation odds, while there is no effect for debates held in the Commons Chamber.

variability in the posterior – particularly when parameters are correlated – implying that in some cases the estimated standard errors may be too small.

[18] Estimates for the additional coefficients described in equation 4 are given in appendix table S6.

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**	
$\varphi_{\textbf{Margin}}$	-0.3	-0.28	-0.24	-0.25	-0.41	-0.02	-0.56
	(-0.65, 0.02)	(-0.59, 0.03)	(-0.5, 0.04)	(-0.54, 0.02)	(-0.64, -0.15)	(-0.43, 0.41)	(-0.95, -0.18)
$\varphi_{\textbf{Government}}$	-0.29	-0.5	-0.25	-0.31	-0.27	0.04	-0.59
	(-0.53, -0.07)	(-0.68, -0.32)	(-0.43, -0.06)	(-0.47, -0.14)	(-0.45, -0.11)	(-0.26, 0.34)	(-0.83, -0.34)
$\varphi_{\textbf{New MP}}$	0.16	0.16	0.2	0.21	0.09	0.06	0.17
	(0, 0.3)	(0.01, 0.32)	(0.06, 0.34)	(0.05, 0.35)	(-0.03, 0.21)	(-0.15, 0.28)	(-0.05, 0.4)
φ_0	0.1	0.21	0.25	0.29	0.25	0.01	0.32
	(0.03, 0.17)	(0.15, 0.29)	(0.14, 0.35)	(0.22, 0.35)	(0.18, 0.32)	(-0.1, 0.12)	(0.22, 0.43)
Debate location	C+W	C+W	C+W	C+W	C+W	C	W
MP random effects	×	✓	×	✓	✓	✓	
Debate random effects	×	×	✓	✓	✓	✓	
MP-topic random effects	×	×	×	×	✓	✓	
N parameters	667	1332	3195	3860	17080	17745	
AIC	472371.2	445072.6	441118.4	414088.8	392933.2	394475.3	

Table 1: **Second-level φ effects (debate participation)**
Median estimates and associated 95% posterior intervals for the second-level φ coefficients described in equation 3.

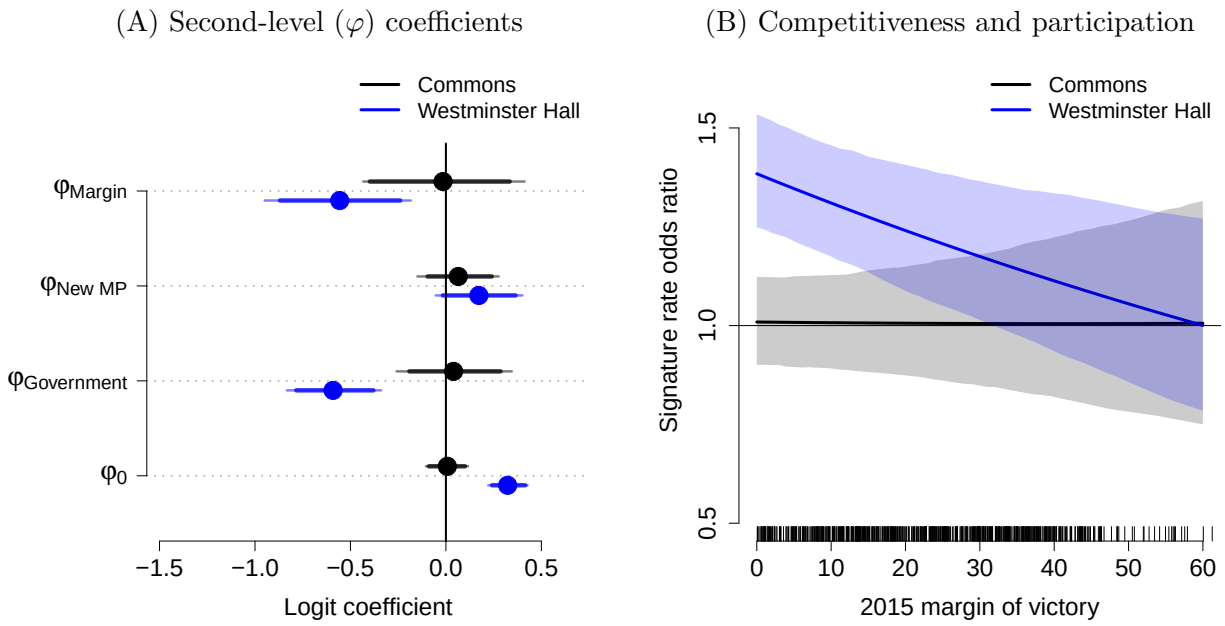

Figure 3: **Effects of signature rate on debate participation**
Panel A: median estimates and 90% (thin lines) and 95% (thick lines) posterior intervals for the φ coefficients from model 6 in table 1. Panel B: Effects of the signature rate on participation (odds-ratios) across the range of $Margin$ for debates in Westminster Hall (blue) and the Commons' Chamber (black).

To further explore the heterogeneity in these effects, I plot the second-level φ coefficients for model 6, with associated 90 and 95% posterior intervals, in panel A of figure 3. Blue points are Westminster Hall coefficients, and black points represent coefficients relating to the Commons' Chamber. Consistent with hypothesis 2, the plot shows that – regardless of MP characteristics – the signature-rate has no effect on participation for debates held in the main chamber of the Commons. By contrast, for debates in Westminster Hall, there are important interaction effects between MP characteristics and the signature-rate. For example, in line with hypothesis 4, the estimated $\varphi_{\text{Government}}$ coefficient is negative, implying that the effect of petition signatures is smaller for MPs who hold government positions than it is for backbench MPs.

Similarly, the negative coefficient for φ_{Margin} also provides support for hypothesis 3, and suggests that the effect of petition signatures is larger for MPs from more competitive constituencies. Panel B of figure 3 depicts the effects of the signature-rate on participation (as

an odds-ratio) across the range of the Margin variable. For debates held in Westminster Hall (blue line), the effect of petition signatures is strongest for MPs in competitive constituencies, and weakest for MPs in very safe seats. The substantive magnitude of this effect is non-trivial: increasing the signature rate in a competitive constituency (Margin = 1) from half of the national rate to 5 times the national rate leads to a 5 percentage point increase in the probability of participating in debate,[19] relative to a baseline probability of just 3 percent. However, the same increase in the signature rate for uncompetitive constituencies (Margin = 60) results in no increase in participation probability. For debates in the Commons Chamber, by contrast, the effect of signatures is zero regardless of an MP's electoral security. Finally, although the $\varphi_{\text{New MP}}$ coefficient is positive, it is not consistently significant across model specifications, suggesting only limited support for hypothesis 5.

In sum, the results here suggest that even conditioning on the topics under consideration, MPs are responsive to constituent activism as expressed through e-petitions, but also that this responsiveness is clearly conditioned by the strength of party discipline and by the electoral incentives an MP faces.

Petition agreement

For those MPs who do participate in petition debates, is there a relationship between signatures and petition *agreement*? If e-petitions are effective, we would expect MPs who receive many signatures from their constituents to actively support a petition during the course of debate. I investigate this question by focussing on the subset of 487 MPs who participated in the 46 petition debates in the sample. An MP is coded as being in agreement with a petition if she expressed overt support for the proposal given in the petition text.[20] In this section, the dependent variable therefore measures whether an MP expressed agreement with a given e-petition ($y_{id} = 1$) or not ($y_{id} = 0$).

[19] Based on MPs who do not hold a government position and who are not new to the house.

[20] The author and an RA read all contributions to each petition debate, and coded whether each MP was "For" or "Against" the motion in the petition (or "Don't know"). In cases where more than one petition was discussed in a given debate, coding was based on agreement with the largest petition. "Don't know" responses are coded as "Against". Coders agreed on the speech codings in 90 % of cases. The results in table 2 are based on the author's coding, but the secondary codings result in substantively and statistically similar results (appendix table S8).

As with the previous analysis, I model MP agreement using a hierarchical logit model which includes MP-specific signature effects (γ_i), MP intercepts (α_i), and debate intercepts (δ_d) at the first level. Here, the MP effects capture the tendency for MPs to agree with petitions in general, and the debate effects capture whether some petitions are more agreeable to all MPs than others. Predictors at the second-level of the model are identical to those given in equation 3: I model the MP-signature effects (γ_i) as a function of whether the MP is new to the House, whether the MP held a government position, and the electoral margin of the MP.[21]

In contrast to the analysis above, I am unable to estimate the MP-topic random effects (β_{ik}) here. To do so would require coding each MP's *position*, rather than merely their participation, on each of the non-petition debates discussed in parliament. Manual coding of this sort is infeasible for such a large number of debates, and traditional scaling approaches are not well-suited to a system with strong party discipline and strategic voting (Spirling & McLean 2006). Accordingly, because this secondary analysis is potentially prone to MP-specific topic confounding, the results should be considered as illustrative. Given the smaller data and simpler model, I estimate the model with MCMC in Stan.

Results

The estimates for the second-level φ_i coefficients from the approval analysis are presented in table 2. As before, the estimates of the second-level coefficients are relatively stable across the different specifications, though, due to the much smaller sample size, they are also somewhat noisier.

In general, this analysis suggests that in addition to the effects on debate participation, petition signatures also matter for MP approval. The baseline signature-effect parameter, φ_0, is positive in all model specifications and significant in all cases except for the Westminster Hall coefficient in model 5. Based on the estimates in model 4, for a non-government MP who is not new to the house, and has the mean level of electoral competitiveness, increasing the signature rate from half of the national rate to 5 times the national rate leads to a 42

[21] Again, I include the same predictors in the second-level model for the α_i parameters (appendix table S7).

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	
$\varphi_{\textbf{Margin}}$	0.58	0.59	0.03	0.4	0.8	-0.45
	(-0.72, 1.76)	(-1.22, 1.94)	(-1.35, 1.21)	(-1.38, 2.03)	(-1.01, 3.28)	(-3.57, 2.3)
$\varphi_{\textbf{Government}}$	-1.32	-1.53	-1.46	-1.73	0.09	-1.97
	(-2.27, -0.26)	(-2.99, -0.14)	(-2.47, -0.25)	(-3.85, -0.06)	(-2.39, 2.84)	(-4.65, 0.7)
$\varphi_{\textbf{New MP}}$	0.06	0.01	0.04	0.09	0.37	0.05
	(-0.3, 0.36)	(-0.5, 0.49)	(-0.34, 0.41)	(-0.45, 0.69)	(-0.47, 1.11)	(-0.81, 0.98)
φ_0	0.51	0.75	0.6	0.68	0.76	0.68
	(0.15, 1.02)	(0.29, 1.27)	(0.23, 1.1)	(0.09, 1.37)	(0.09, 1.56)	(-0.2, 1.58)
Debate location	C+W	C+W	C+W	C+W	C	W
MP random effects	×	✓	×	✓	✓	
Debate random effects	×	×	✓	✓	✓	
N parameters	494	986	542	1034	1525	
AIC	2760.4	3352.1	2646.4	3208.3	4146.3	

Table 2: **Second-level φ effects (petition approval)**
Median estimates and associated 95% posterior intervals for the second-level φ coefficients for the petition approval model.

percentage point increase in the probability of petition approval. Conditional on participating in petition debate, MPs are substantially more likely to express support for a petition in their speeches when many of their constituents have signed that petition.

There is, however, less evidence of heterogeneity here. We can draw few conclusions from model 5 which separates the signature effects by debate location, as almost all the second-level coefficients are imprecisely estimated because of the smaller sample size. We can also see that across all models in table 2, the φ_{Margin} coefficient is indistinguishable from zero: petition signatures are no more or less effective at driving MP support for a petition at high or low levels of electoral security. In addition, the φ_{NewMP} coefficient is also not significantly different from zero in all specifications.

By contrast, the $\varphi_{\text{Government}}$ coefficient is negative across all specifications, and is significantly different from 0 in all models except model 5, suggesting that petition signatures are largely inconsequential for the positions taken by government ministers. As ministers appear in petition debates to present the government's response to petitioner concerns, it is unsurprising that the positions these MPs take are unresponsive to the particular views of their own constituents.

In sum, while these approval results are potentially subject to the topic-confounding concerns explained in previous sections, the evidence is broadly consistent with the notion that MPs use petition debates to demonstrate responsiveness to constituent concerns. Combined with the participation results, this suggests that e-petitions provide useful information about constituent preferences, and that this information can strengthen the dyadic relationship between voters and MPs in Westminster.

Alternative explanations

One concern with the previous analyses is that the relationship between signatures and debate behaviour may be attributable to other factors. First, rather than being a tool for constituents to communicate their preferences to legislators, petitions may instead be a tool for legislators to mobilise public support for their own favoured projects. However, there are two reasons why this reverse-causality argument is not a serious concern here. First, in contrast to traditional

petitions in the Commons, e-petitions do not require the support or 'sponsorship' of an MP before they are considered for debate, and previous work shows that e-petitions are normally promoted by individual citizens with little help from MPs themselves (Wright 2015*b*). Second, the vast majority of traffic to the website that hosts e-petitions comes via social media (Yasseri et al. 2017, 11). If MPs encourage their constituents to sign particular petitions, this should be obvious from their activity on social media platforms. In appendix section S3, I show that very few MPs use social media to promote e-petitions, making it unlikely that the results presented above are attributable to MP activism.

A second possibility is that some third party may independently lobby politicians to take action on a certain issue, and also try to garner public support for that issue. If an interest group simultaneously mobilises local support for a campaign while also making representations to legislators, the relationship between constituent activism and legislative behaviour may therefore also be confounded. However, the topic-control strategy implies that any alternative explanation must account for the fact that MPs behave differently in debates on specific, locally-popular petitions than they do in other debates that relate to the same issues as those raised in those petitions. If interest groups pressure MPs to take action on an issue *in general*, then MPs will respond by devoting attention to that issue across many debates and this will be captured by the MP-topic coefficients. It is only if interest groups mobilise both politicians and voters on a specific petition that this is a concern for the results presented here.

As with any observational analysis, we may have remaining concerns regarding alternative mechanisms, but any account of these findings must explain the change in behaviour on petition debates relative to other debates on similar topics, the fact that these effects only affect MPs from marginal constituencies, and the fact that the effects only hold for debates that are not subject to strong party discipline.

Conclusion

On June 24th 2016, MPs in Westminster Hall debated a petition calling on the government to "Prevent Donald Trump from making a State Visit to the United Kingdom" which received nearly 2 million signatures. The debate was very well attended, so much so that the sitting was

extended to accommodate the increased demand for floor time. Although the government opposed the petition's aims, MPs from across the political spectrum spoke in favour of cancelling the state visit. In the final speech of the debate, Paul Flynn MP concluded: "This has been an extraordinary event... We are expressing the voice of the people and a thunderous voice it has been... we are reacting to the voice of the people, and to the anger and fear outside. It is a good day for Parliament." (Flynn 2017) The main findings here suggest that this is more than simple hyperbole. I have shown that, in general, when constituent opinion as expressed via an e-petition is strong, MPs are more likely to lend their voice to the expression of that opinion in parliamentary debates. These results are therefore encouraging, as they suggest that providing voters with new mechanisms for communicating their political preferences can lead to stronger representational ties between citizens and politicians.

However, the conditional nature of these effects should also serve as a warning of the limits of these types of policy innovations. Notably, the representational benefits of e-petitions are largely absent on policy issues that are marked by pre-existing partisan conflict and strong party discipline, and the effectiveness of online activism is clearly conditioned by incentives created by the electoral system. Accordingly, while some see the internet as a force which has the potential to dramatically alter representational relationships in politics (Coleman & Blumler 2009), the findings here suggest that online political activism is more likely to shape political behaviour at the margins and will likely be constrained by existing institutional structures.

These findings also speak to a growing empirical literature on dyadic representation in parliamentary systems. The combination of an SMD electoral system and a parliamentary government creates countervailing incentives for MPs: be a loyal party member and improve the party brand, or respond faithfully to constituent demands and increase electoral security by building a strong personal vote. While party discipline is clearly a primary determinant of MPs' *voting* decisions in these systems, the findings here highlight that responsiveness to voter preferences via other forms of legislative behaviour is non-zero. Whether they navigate the party-voter trade off by making use of roll-call votes that are not subject to party discipline (Hanretty et al. 2016), by internal lobbying of party leaders (Butler et al. 2017),

or, as demonstrated here, through the strategic use of parliamentary speech (Soroka et al. 2009, Blidook & Kerby 2011), it is clear that MPs are aware of and respond to constituent preferences with many different types of legislative behaviour.

There are many other dimensions on which we might want to evaluate the effectiveness of online activism. A common criticism of e-petition systems, for example, is that even popular petitions have little tangible impact on the policy-making process (Wright 2015*a*). For instance, despite the vocal opposition of many MPs in the Trump debate, the US President did indeed visit the UK in mid-2018. However, there are some instances where e-petitions clearly played a role in promoting policy change (Miller 2008), and more work is need to assess what (if any) downstream consequences the behavioural effects documented here have for policymaking. Future research might also address whether e-petitions help cultivate further interest in politics, and whether they encourage signatories to participate in politics in other ways.

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Appendix

S1. Descriptive analysis of petition signatures

Table S1 presents the results for a simple descriptive analysis which documents the variation in aggregate petition signature rates across parliamentary constituencies in the UK. I calculate the total number of signatures per constituency across all petitions in the data that received over 1000 signatures nationally. I then divide this by the number of people living in the constituency in 2015 to get the aggregate number of signatures per capita, and use this as an outcome variable in a linear regression model. The right-hand side variables are demographic – median pay; percentage female; percentage under 30; population density; percentage of students; percentage working in industry; percentage of home owners; percentage white; percentage unemployed – or political – vote for 'Leave' in the 2016 UK-EU referendum; Labour, Liberal Democrat, SNP, and 'Other' vote share in 2015. In model 2 of table S1, I also include regional fixed-effects.

The table shows that the volume of signatures that constituencies contribute across petitions vary in predictable ways. Constituencies which have high numbers of young people, lower levels of industrial workers, lower levels of unemployment and which are less ethnically diverse tend to contribute more signatures to e-petitions. Similarly, constituencies in which more people supported 'Remain' in the EU referendum are significantly more likely to sign e-petitions. These findings are consistent on petition-signing in Germany, where Lindner & Riehm (2011), for example, find that individuals who sign e-petitions are younger and more educated, on average, than the general population.

Table S1: Constituency-level determinants of signature rates

	Signatures per capita	
	(1)	(2)
Median pay	0.0001	−0.001
	(0.001)	(0.001)
Female	0.004	0.0001
	(0.004)	(0.004)
Young	0.007***	0.006***
	(0.002)	(0.002)
Pop. Density	0.001***	0.001***
	(0.0002)	(0.0002)
Students	−0.004***	−0.003**
	(0.001)	(0.001)
Industry	−0.005***	−0.003***
	(0.001)	(0.001)
Home Ownership	−0.0001	0.001
	(0.001)	(0.0005)
White	0.002***	0.002***
	(0.0002)	(0.0003)
Unemployed	−0.017***	−0.008***
	(0.003)	(0.003)
Leave Share	−0.514***	−0.643***
	(0.038)	(0.038)
Labour 2015	−0.011*	0.013*
	(0.007)	(0.007)
LibDem 2015	−0.027	0.005
	(0.020)	(0.018)
SNP 2015	−0.165***	−0.042
	(0.012)	(0.030)
Other 2015	0.015	0.056**
	(0.025)	(0.023)
Constant	0.438**	0.506***
	(0.199)	(0.179)
Regional Fixed-Effects	No	Yes
Observations	631	631
R^2	0.685	0.753
Note:	*p<0.1; **p<0.05; ***p<0.01	

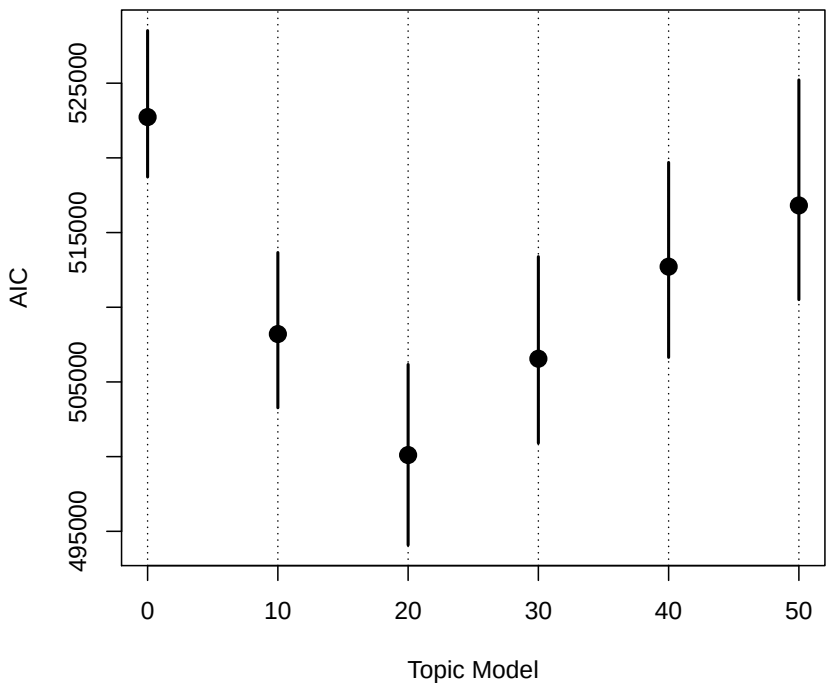
Title	Signatures	Date
EU Referendum Rules triggering a 2nd EU Referendum	4150262	Sep 2016
Prevent Donald Trump from making a State Visit to the United Kingdom.	1863707	Feb 2017
Give the Meningitis B vaccine to ALL children, not just newborn babies.	823349	Apr 2016
Block Donald J Trump from UK entry	586930	Jan 2016
Accept more asylum seekers and increase support for refugee migrants in the UK.	450287	Sep 2015
Donald Trump should make a State Visit to the United Kingdom.	317542	Feb 2017
Make the production, sale and use of cannabis legal.	236995	Oct 2015
Stop spending a fixed 0.7 per cent slice of our national wealth on Foreign Aid	235979	Jun 2016
To debate a vote of no confidence in Health Secretary the Right Hon Jeremy Hunt	231136	Sep 2015
Vote no on military action in Syria against IS in response to the Paris attacks	227745	Dec 2015
STOP CAMERON spending British taxpayers money on Pro-EU Referendum leaflets	221866	May 2016
Stop allowing immigrants into the UK.	216949	Oct 2015
No more school penalty fines and bring back the 10 day authorised absence	204790	Jul 2016
Make fair transitional state pension arrangements for 1950s women	193185	Feb 2016
No UK airstrikes on Syria.	190223	Dec 2015
Put a max of 1200 on car insurance for 18-25 year olds	185175	Mar 2017
Keep the NHS Bursary	162568	Jan 2016
Introduce a tax on sugary drinks in the UK to improve our children's health.	155516	Nov 2015
Make it illegal for a company to require women to wear high heels at work	152420	Mar 2017
Close all retail on boxing day, retail isn't needed on boxing day!	148327	Dec 2016
Stop retrospective changes to the student loans agreement	134194	Jul 2016
Free childcare when both parents are working. Not just those who are on benefits	133921	Nov 2016
Prevent the scrapping of the maintenance grant.	133069	Jan 2016
Invoke Article 50 of The Lisbon Treaty immediately.	128238	Oct 2016
Give status to Police Dogs and Horses as 'Police Officers'	127729	Nov 2016
April's Law	127296	Mar 2017
Make an allowance for up to 2 weeks term time leave from school for holiday.	127199	Oct 2015
Ban driven grouse shooting	123077	Oct 2016
Jeremy Hunt to resume meaningful contract negotiations with the BMA.	120680	Mar 2016
Fund more research into brain tumours, the biggest cancer killer of under-40s	120129	Apr 2016
Make it a specific criminal offence to attack any member of NHS Medical Staff.	118056	Feb 2017
Force child cancer to the forefront of the NHS and government funding schemes.	117183	Nov 2016
Scrap the 35k threshold for non-EU citizens settling in the UK	114636	Mar 2016
Scrap plans forcing self employed and small business to do 4 tax returns yearly	114504	Jan 2016
Authorise open book examinations for GCSE English Literature 2017	110876	Apr 2017
The DDRB's proposals to change Junior Doctor's contracts CANNOT go ahead.	110065	Oct 2015
Shut down the domestic ivory market in the UK.	108530	Feb 2017
End the badger cull instead of expanding to new areas	108320	Mar 2017
Demand an end to the pay restraint imposed on agenda for change NHS staff.	106409	Jan 2017
Debate in the House the Local Government Pension Scheme Investment Regulations	105772	Oct 2016
Restrict the use of fireworks to reduce stress and fear in animals and pets	104038	Jun 2016
Include expressive arts subjects in the Ebacc	102499	Jul 2016
Urge the South Korean Government to end the brutal dog meat trade	102131	Sep 2016
Ban all non-recyclable/non-compostable packaging in the UK	101941	Jan 2017
Don't kill our bees! Immediately halt the use of Neonicotinoids on crops	99909	Dec 2015
Don't ban Trump from the United Kingdom	46622	Jan 2016
Free Sergeant Alexander Blackman	34440	Sep 2015
Repeal the 1972 European Communities Act immediately on EU Referendum OUT vote.	32943	Oct 2016
Ask the Government not to invoke article 50 of the Lisbon treaty.	27151	Oct 2016
Let Parliament decide whether or not we remain a member of the European Union	25729	Oct 2016
Protect grouse Moors and grouse shooting	25322	Oct 2016
Not to allow freedom of movement as part of any deal with the EU after Brexit	21537	Oct 2016
The 23 June should be designated as Independence Day, and celebrated annually.	21292	Oct 2016
Save British Steel making. Scunthorpe, Teesside, Port Talbot etc.	18429	Oct 2015
Stop the scathing cuts to the Police budget.	9947	Nov 2015
Stop the destructive 'building our future' office closure programme in HMRC	2585	Nov 2015

Table S2: Debated e-petitions

S2. Topic control strategy validation

I validate the topic control strategy in three ways. First, if MPs speak predictably on certain issue areas, then models that include topics should have better predictive performance than models without this information. In figure S1 I show that including MP-specific topic effects clearly decreases the AIC of the model relative to a null model which omits these effects. Furthermore, the model with 20 topics performs best in terms of AIC, and I focus on the results from this model throughout the paper. Figure S2 presents the coefficients of interest from all 5 models. Regardless of the specific topic model, the results are statistically and substantively very similar.

Figure S1: AIC by topic



Second, table S3 lists the names of 6 MPs with the highest β_{ik} coefficients from 4 example topics. The results are encouraging as MPs are associated with topical areas on which we have strong intuitions that they should be active. For example, the listed MPs with the highest coefficients for the "eu;european;union;referendum;negoti" topic are all well-known campaigners on the issue of the relationship between the UK and the EU and include MPs from both the 'Remain' (Hillary Benn, Stephen Gethins) and 'Leave' (John Redwood, William Cash, David Davis) camps. Similarly, the MPs appearing in the "nhs;hospit;polic;fire;doctor" topic also make sense in substantive terms: Jeremy Hunt, Alistair Burt, and Ben Gummer were the Secretary, Minister, and Undersecretary of State for Health, respectively, during the 2015-2017 parliament, and Heidi Alexander was the shadow Secretary of State for Health.

Similar patterns hold for the MPs listed under the "arm;war;veteran;militari;personnel"

Table S3: MPs with highest β_{ik} coefficients by topic

eu;european;union;referendum;negoti	**nhs;hospit;polic;fire;doctor**
John Redwood	Alistair Burt
Mr Pat McFadden	Mike Penning
Sir William Cash	Mr Jeremy Hunt
Mr David Davis	Ben Gummer
Hilary Benn	Heidi Alexander
Stephen Gethins	Dr Sarah Wollaston
arm;war;veteran;militari;personnel	**school;student;educ;children;parent**
Mark Lancaster	Nick Gibb
Bob Stewart	Edward Timpson
Brendan O'Hara	Mr Sam Gyimah
Kirsten Oswald	Nicky Morgan
Sir Julian Brazier	Priti Patel
Sir Michael Fallon	Lucy Powell

topic. During the relevant parliament, Michael Fallon, Julian Brazier, and Mark Lancaster were the Secretary of State, Minister, and Undersecretary for Defence, respectively, Brendan O'Hara was a Shadow spokesperson for Defence, and Bob Steward was a member of the Defence select committee. Finally, MPs who feature on the "school,student,educ,children,parent" topic held government roles in the Department for Education (Nicky Morgan, Nick Gibb, Edward Timpson, Sam Gyimah) or were members of the opposition front-bench focussed on education (Lucy Powell). Overall, these tables reveal that the text-based control strategy I employ has reassuring face validity: the procedure automatically associates MPs with topical areas on which we have strong intuitions to believe they should be active.

Finally, the control strategy relies on how well we can approximate the topics of petition debates in the full sample of non-petition debates. To investigate this, table S4 shows, for each of the petition debates in the data, the titles of the three closest matching *non-petition* debates according to the cosine-similarity between the topic vectors.[22] Again, the results are reassuring as petition debates are matched with non-petition debates that deal with very similar policy areas. For example, the petition debate on the "Refugee Crisis in Europe" is matched to debates on the "Refugees and Counter-terrorism", and "Syria". Similarly, the petition debate on the "EU Referendum Leaflet" is matched to other debates on the EU Referendum and Europe. Indeed, across almost all petition debates in the sample, the topic model clearly clusters petition debates with non-petition debates that focus on substantively very similar policy areas.

Linear regression estimates (table S5) show that a five-unit increase in the signature rate is associated with an increase of 14 percentage points in participation probability of petition debates. However, a five-unit increase in the signature rate is also associated with a 7 percentage point increase in participation in these closest matching *non-petition* debates. This reinforces the importance of including MP-specific topic effects: MPs are likely to be active on topics that are of greater interest to their constituents, even in the absence of the information

[22] I calculate the cosine similarity between the topic vectors of each petition debate and every other non-petition debate in the data. I then rank the debates for each petition debate according to this metric and select the top three to display in table S4.

provided by an e-petition.

Overall, the topic-control strategy improves our ability to predict MP debate participation; associates MPs to topics in ways that are consistent with intuitions about British politics; and links petition-debates to counterfactual non-petition-debates that are very similar in terms of their topical content.

Table S4

Petition Debate	Closest Matching Debates
Refugee Crisis in Europe	Syria: Refugees and Counter-terrorism; Rohingya Community; Burundi
NHS (Contracts and Conditions)	NHS Reform; Emergency Medicine; North East Ambulance Service
Sgt Alexander Blackman (Marine A)	Battle of Jutland Centenary; Clydebank Blitz Anniversary; Centenary of the Battle of the Somme
Cannabis	Coeliac Disease and Prescriptions; ONeill Review; Alcohol Harm
Immigration	Europe, Human Rights and; Human Rights Act; Gibraltar and Spain
Term-time Leave	Child Food Poverty; School Penalty Fines and Authorised Absence; Term-time Holidays
Junior Doctors Contracts	Junior Doctors Contracts; Hospitals in Special Measures; Mid Yorkshire Hospitals NHS Trust
Steel Industry	UK Steel Industry; Hinkley Point C; Energy Spending Priorities: Investors
Policing	Police Funding, Crime and Community Safety; Police Grant Report; Safer Neighbourhood Policing: London
HMRC Office Closures	HMRC: Building our Future Plan; HMRC Estate; HMRC Regional Centres
Sugary Drinks Tax	Childhood Obesity Strategy; Badger Culling/Bovine TB; Badger Culling
ISIL in Syria	Libya; Aleppo/Syria: International Action; Kashmir
Neonicotinoids on Crops	Lead Shot Ammunition; Welfare of Young Dogs Bred for Sale; Exotic Pets Trade
NHS Bursary	Student Nursing; NHS Bursaries; Children and Young Peoples Mental Health
Donald Trump	Stormont; Hate Speech; Northern Ireland (Ministerial Appointments
Student Maintenance Grants	Child Poverty in the; Education; Child Poverty
Small Businesses: Tax Reporting	HMRC Penalty Surcharges; Multinational Enterprises; Small Businesses: Late Payments
Transitional State Pension Arrangements for Women	Implementation of the 1995 and 2011 Pensions Acts; State Pension Age Equalisation; Implementation of the 1995 and 2011 Pension Acts
Non-EU Citizens: Income Threshold	EU Referendum: Immigration and; Benefit Entitlement Bill; EU Migrants: National Insurance Numbers
BMA (Contract Negotiations)	Junior Doctors Contract; Junior Doctors Contracts; Junior Doctors Contract Negotiations
Brain Tumours	Fibromyalgia; Cough-Assist Machines; Bowel Cancer Screening Age
Meningitis B Vaccine	Cochlear Implantation; Alcohol Consumption Guidelines; Enriched Culture Medium Test
EU Referendum Leaflet	EU Referendum Rules; EU Referendum; Europe: Renegotiation
Fireworks	Welfare of Young Dogs Bred for Sale; Snares; Driven Grouse Shooting
Foreign Aid Expenditure	International Development; Syrian Refugee Crisis; Refugees in Calais
EBacc: Expressive Arts Subjects	Childrens Creative and Cultural Experiences; Access to the Arts; Worcester University Arena: Disabled
School Penalty Fines and Authorised Absence	Term-time Leave; Term-time Holidays; Child Food Poverty
Student Loans Agreement	Disability Benefits; Student Loan Repayments; Child Benefit
EU Referendum Rules	European Union Bill; Parliamentary Sovereignty and EU Renegotiations; The Government's Plan for Brexit
UK Exit from the European Union	Parliamentary Scrutiny of Leaving the EU; Article 50; Joint Ministerial Committee
Local Government Pension Scheme	Local Government Pension Scheme; Older Carers; Older Women: Caring Support
Driven Grouse Shooting	Exotic Pets Trade; Snares; Lead Shot Ammunition
Police Dogs and Horses	Dog Theft: Sentencing; Animal Cruelty: Sentencing; Sentencing
Free Childcare	Access to Local Schools; Childcare Bill [Lords]; Disabled People: Work
Child Cancer	Diabetes-related Complications; Young-onset Parkinsons Disease; Diabetes Technologies
Retail Store Closure: Boxing Day	Serious Fraud Office: Bryan Evans; Faulty Tumble Dryers; Basic Payment Scheme
Non-recyclable and Non-compostable Packaging	Microbead Ban; Food Waste; Faulty Electrical Imports
Agenda for Change: NHS Pay Restraint	Agency Staffing Expenditure; NHS: Financial Performance; National Health Service
Domestic Ivory Market	Backbench Business; Dog Meat; Dog Meat Trade
President Trump: State Visit	London Attacks; Donald Trump; Changes in US Immigration Policy
Attacks on NHS Staff	Police Officer Safety; Child Sexual Abuse Cases: Metropolitan Police; Departmental Absence Management
Workplace Dress Codes (High Heels)	Pension Freedoms; Draft Exclusivity Terms in; Guaranteed Income for Retirees
Child Abuse Offences (Sentencing)	Court System; Knife Crime; Prisons and Courts Bill
Car Insurance: Young People	Vehicle Speed Outside Schools; Gypsies and Travellers; National Road Safety Targets
Badger Culling	Badger Culling/Bovine TB; Sugary Drinks Tax; Badger Cull
GCSE English Literature Exams	Ofsted Inspections; Compulsory Emergency First Aid; Technical and Further Education

Table S5

	Participation	
	Petition Debates	Closest matching non-petition debates
	(1)	(2)
Relative Signature Rate	0.029***	0.013***
	(0.002)	(0.002)
Constant	0.046***	0.076***
	(0.001)	(0.002)
Observations	29,861	29,861
R^2	0.008	0.001
Note:		*p<0.1; **p<0.05; ***p<0.01

Table S6: Second-level predictors (debate participation)

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**	
$\varphi_{\textbf{Margin}}$	-0.3	-0.28	-0.24	-0.25	-0.41	-0.02	-0.56
	(-0.65, 0.02)	(-0.59, 0.03)	(-0.5, 0.04)	(-0.54, 0.02)	(-0.64, -0.15)	(-0.43, 0.41)	(-0.95, -0.18)
$\varphi_{\textbf{Government}}$	-0.29	-0.5	-0.25	-0.31	-0.27	0.04	-0.59
	(-0.53, -0.07)	(-0.68, -0.32)	(-0.43, -0.06)	(-0.47, -0.14)	(-0.45, -0.11)	(-0.26, 0.34)	(-0.83, -0.34)
$\varphi_{\textbf{New MP}}$	0.16	0.16	0.2	0.21	0.09	0.06	0.17
	(0, 0.3)	(0.01, 0.32)	(0.06, 0.34)	(0.05, 0.35)	(-0.03, 0.21)	(-0.15, 0.28)	(-0.05, 0.4)
φ_0	0.1	0.21	0.25	0.29	0.25	0.01	0.32
	(0.03, 0.17)	(0.15, 0.29)	(0.14, 0.35)	(0.22, 0.35)	(0.18, 0.32)	(-0.1, 0.12)	(0.22, 0.43)
$\mu_{\textbf{Margin}}$		-0.61		-0.67	-0.95	-0.84	
		(-0.87, -0.36)		(-0.9, -0.46)	(-1.07, -0.82)	(-0.99, -0.71)	
$\mu_{\textbf{Government}}$		-0.56		-0.58	-0.95	-0.94	
		(-0.69, -0.44)		(-0.73, -0.43)	(-1.05, -0.86)	(-1.05, -0.82)	
$\mu_{\textbf{New MP}}$		0.27		0.24	0.31	0.33	
		(0.18, 0.37)		(0.12, 0.34)	(0.25, 0.38)	(0.27, 0.39)	
μ_0		-0.49		-0.49	-1.12	-0.73	
		(-0.56, -0.41)		(-0.54, -0.44)	(-1.16, -1.08)	(-0.76, -0.69)	
$\phi_{\textbf{Petition}}$			0.2	0.17	0.29	0.14	
			(-0.01, 0.4)	(-0.05, 0.37)	(0.06, 0.52)	(-0.1, 0.39)	
ϕ_0			-1.01	-0.57	0.62	-0.79	
			(-1.05, -0.98)	(-0.59, -0.55)	(0.58, 0.65)	(-0.83, -0.75)	
Debate location	C+W	C+W	C+W	C+W	C+W	C	W
MP random effects	×	✓	×	✓	✓	✓	
Debate random effects	×	×	✓	✓	✓	✓	
MP-topic random effects	×	×	×	×	✓	✓	
N parameters	667	1332	3195	3860	17080	17745	
AIC	472371.2	445072.6	441118.4	414088.8	392933.2	394475.3	

Note: Table presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equations 3 and 4 for the debate participation analysis. Model 1 is a baseline model, models 2 and 3 introduce MP and debate random-effects respectively, and model 4 includes both sets of random effects. Model 5 includes the matrix of MP-topic random-effects (β_{ik}). Model 6 includes the full set of random effects, and also estimates the effect of the signature rate on debate participation for each MP ($gamma_{i,l}$) in two locations: the Commons (C) and Westminster Hall (W).

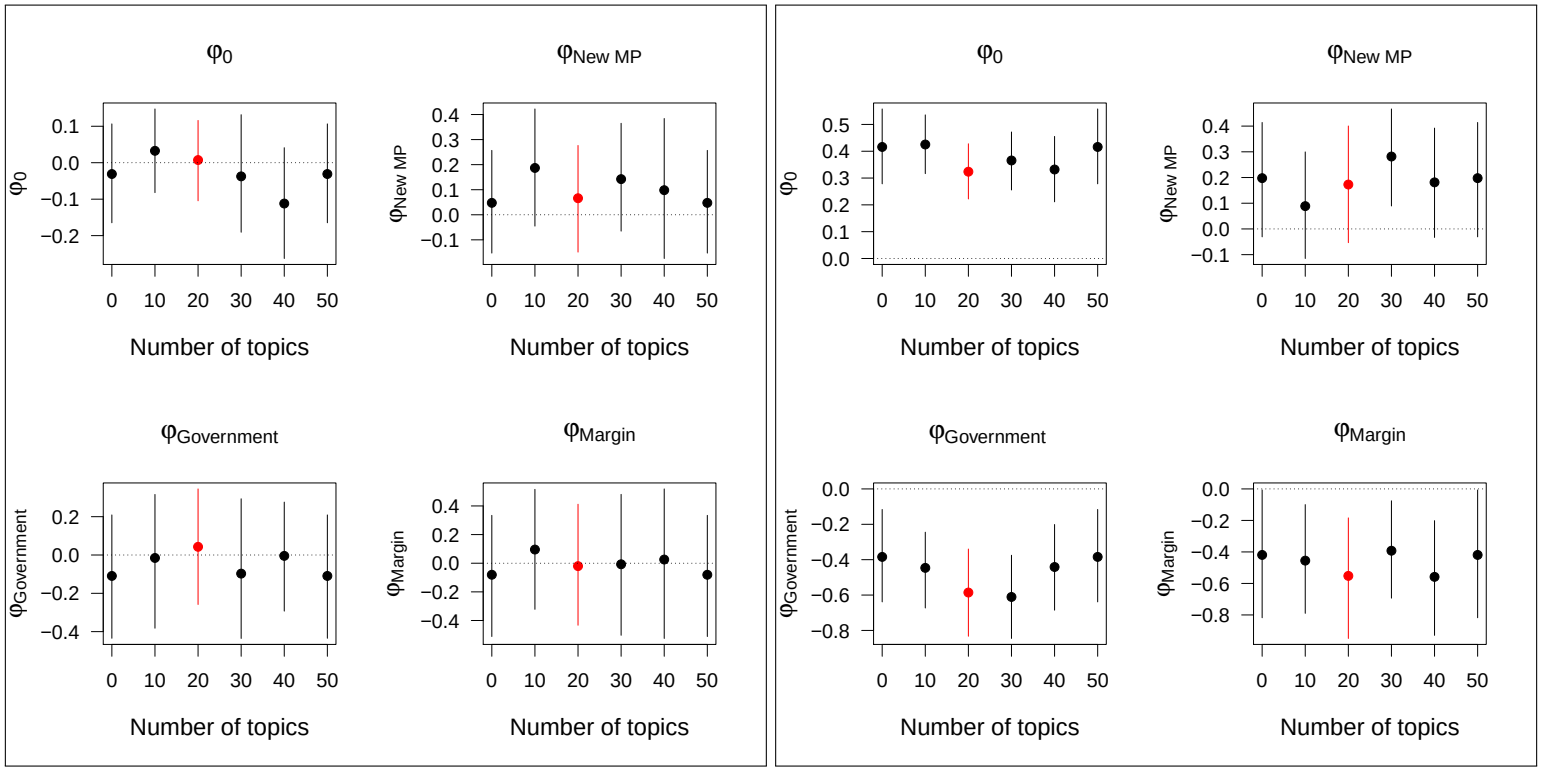
Table S7: Second-level predictors (petition approval)

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	
$\varphi_{\textbf{Margin}}$	0.58 (-0.72, 1.76)	0.59 (-1.22, 1.94)	0.03 (-1.35, 1.21)	0.4 (-1.38, 2.03)	0.8 (-1.01, 3.28)	-0.45 (-3.57, 2.3)
$\varphi_{\textbf{Government}}$	-1.32 (-2.27, -0.26)	-1.53 (-2.99, -0.14)	-1.46 (-2.47, -0.25)	-1.73 (-3.85, -0.06)	0.09 (-2.39, 2.84)	-1.97 (-4.65, 0.7)
$\varphi_{\textbf{New MP}}$	0.06 (-0.3, 0.36)	0.01 (-0.5, 0.49)	0.04 (-0.34, 0.41)	0.09 (-0.45, 0.69)	0.37 (-0.47, 1.11)	0.05 (-0.81, 0.98)
φ_0	0.51 (0.15, 1.02)	0.75 (0.29, 1.27)	0.6 (0.23, 1.1)	0.68 (0.09, 1.37)	0.76 (0.09, 1.56)	0.68 (-0.2, 1.58)
$\mu_{\textbf{Margin}}$		-1.41 (-2.64, -0.2)		-1.12 (-2.44, 0.47)	-1 (-2.5, 0.48)	
$\mu_{\textbf{Government}}$		-2.44 (-3.66, -1.46)		-3.43 (-4.79, -2.18)	-3.4 (-4.68, -2.23)	
$\mu_{\textbf{New MP}}$		0.62 (0.14, 1.02)		0.64 (0.17, 1.14)	0.7 (0.22, 1.21)	
μ_0		-2.1 (-3.17, 0.36)		-2.2 (-4.05, -0.5)	-1 (-2.91, 1.52)	
Debate location	C+W	C+W	C+W	C+W	C	W
MP random effects	×	✓	×	✓	✓	
Debate random effects	×	×	✓	✓	✓	
N parameters	494	986	542	1034	1525	
AIC	2760.4	3352.1	2646.4	3208.3	4146.3	

Note: Table presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equations 3 and 4 for the petition approval analysis. Model 1 is a baseline model, models 2 and 3 introduce MP and debate random-effects respectively, and model 4 includes both sets of random effects. Model 5 includes the matrix of MP-topic random-effects (β_{ik}). Model 6 includes the full set of random effects, and also estimates the effect of the signature rate on petition approval for each MP ($gamma_{i,l}$) in two locations: the Commons (C) and Westminster Hall (W).

Figure S2: Second-level predictors (debate participation), all topic models

(A) Commons Chamber φ coefficients (B) Westminster Hall φ coefficients



Note: **Participation results are insensitive to topic-model specification**
The figure depicts the main second-level coefficients of interest (φ_0, $\varphi_{\text{Government}}$, $\varphi_{\text{New MP}}$ and φ_{Margin}) from equation 3 for each of the topic models I estimate. The x-axis of each sub-panel gives the number of topics, and the y-axis gives the value of the relevant coefficient. Panel A presents coefficient estimates that pertain to debates held in the Commons' Chamber, and panel B presents coefficient estimates that pertain to debates held in Westminster Hall. The estimates from the model selected by AIC, and which are presented in the main body of the text in table 1, are in red. Regardless of the number of topics estimated in the CTM model, the results are substantively and statistically very similar.

S3. MP promotion of e-petitions via Twitter

To investigate the possibility that MPs use social media to drive their constituents to sign e-petitions that address issues on which the MPs are already active, I collect data on the tweets sent by MPs during the 2015-2017 parliamentary term from the Twitter API (https://developer.twitter.com). Not all MPs have twitter accounts, but I was able to identify the accounts of 554 MPs (out of 650) during this term. I collected all tweets sent by each MP between May 2015 and May 2017, resulting in just over 1 million tweets. I search the strings of the tweets for links to the official e-petition website, which is easily identifiable through the search string "`https://petition.parliament.uk/petitions/`".

Although some MPs appear to use twitter to promote e-petitions, the numbers are very small. Of the over 1 million tweets in my sample, only 878 include links to the parliamentary e-petition website. Further, of those MPs who I am able to link to twitter accounts, only 258 ever include a link to the petitions website in their tweets during the study period. Given these small numbers, it seems highly unlikely that MP activism could explain the substantive results described in this paper.

	Model 1	Model 2	Model 3	Model 4	Model 5	
$\varphi_{\textbf{Margin}}$	0.24	0.38	0.41	0.27	1.63	-1.11
	(-0.72, 1.58)	(-0.87, 2.07)	(-0.95, 1.9)	(-1.11, 1.88)	(-0.5, 3.71)	(-3.25, 1.05)
$\varphi_{\textbf{Government}}$	-1.09	-1.26	-1.43	-1.12	-1.2	-1.04
	(-2.19, -0.18)	(-3.16, 0.02)	(-2.47, -0.25)	(-2.7, 0.59)	(-4.42, 1.02)	(-3.29, 0.96)
$\varphi_{\textbf{New MP}}$	-0.05	0.06	-0.03	-0.09	0.01	0.17
	(-0.42, 0.35)	(-0.29, 0.42)	(-0.4, 0.33)	(-0.55, 0.28)	(-0.89, 0.8)	(-0.61, 0.92)
φ_0	0.61	0.69	0.61	0.81	0.78	0.96
	(0.25, 0.98)	(0.26, 1.13)	(0.17, 1.09)	(0.38, 1.36)	(0.07, 1.54)	(0.16, 1.62)
$\mu_{\textbf{Margin}}$		-1.34		-1.08	-0.98	
		(-2.6, -0.21)		(-2.5, 0.32)	(-2.47, 0.58)	
$\mu_{\textbf{Government}}$		-1.66		-2.03	-2.14	
		(-2.55, -0.86)		(-3.1, -1.16)	(-3.18, -1.19)	
$\mu_{\textbf{New MP}}$		0.42		0.5	0.52	
		(0.05, 0.82)		(0.1, 0.97)	(0.05, 0.96)	
μ_0		1.34		1.05	-1.1	
		(0.49, 2.67)		(-1.67, 4.28)	(-3, 1.88)	
Debate location	C+W	C+W	C+W	C+W	C	W
MP random effects	×	✓	×	✓	✓	
Debate random effects	×	×	✓	✓	✓	
N parameters	485	968	533	1016	1498	
AIC	2712.3	3322.8	2605.2	3229.3	4167.5	

Table S8: **Second-level φ effects (petition approval), second-coder**
Median estimates and associated 95% posterior intervals for the second-level φ coefficients for the petition approval model based on coding of speeches by second coder.